

# 04/06/2026 - Matemáticas Discretas I (Ude@)

## Anotaciones importantes sobre el cierre

Temas { 1. Sistemas numéricos (04/06/2026) ✓  
2. Álgebra booleana. Definir

Viernes  
05/06/2026

Lunes festivo  
08/06/2026

8:00 AM - 10:00 AM x

10:00 AM - 12:00 PM

Final: 11/06/2026

Evaluación: - 5% → 3 cuestionarios

- 20% → 10% Taller (opcional)  
→ 10% Parcial

Pendiente: 1. Aviso Ude@ de esto  
2. Parciales 2 y 3 calificados (fin de semana)  
3. Programar clase en zoom para el lunes a las 10 AM.

Ejemplos:

$$3B7_{16} \stackrel{?}{=} N_8$$

①:  $16 \rightarrow 10$       ②:  $10 \rightarrow 8$

$$3B7_{16} \xrightarrow{\text{①}} 951_{10} \xrightarrow{\text{②}} 1667_8$$

① Expansión polinómica

$$\begin{aligned} \textcircled{1} \quad 3B7_{16} &= 3 \times 16^2 + B \times 16^1 + 7 \times 16^0 \\ &= 3 \times 16^2 + 11 \times 16^1 + 7 \times 16^0 \\ &= 3 \times 256 + 11 \times 16 + 7 \times 1 \\ &= 951_{10} \\ &= 951 \end{aligned}$$

$$\begin{array}{r} 16 \\ \times 16 \\ \hline 96 \\ 16 \\ \hline 256 \end{array}$$

② Divisiones sucesivas (8)

$$\begin{array}{r} 951 \overline{) 8} \\ 118 \phantom{00} \\ \underline{71} \phantom{00} \\ 38 \phantom{00} \\ \underline{32} \phantom{00} \\ 6 \phantom{00} \\ \underline{6} \phantom{00} \\ 0 \phantom{00} \\ 1 \phantom{00} \\ \underline{8} \\ 0 \end{array}$$

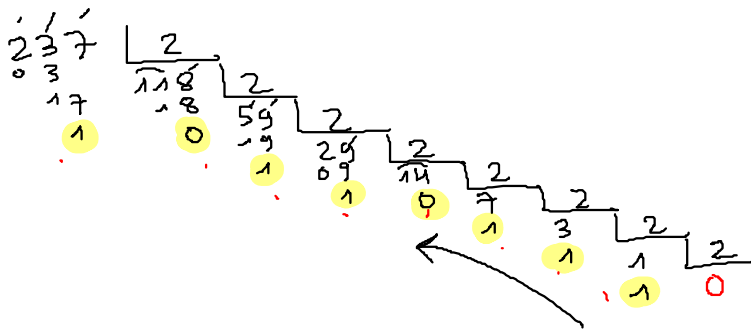
Solución:

$$3B7_{16} = 1667_8$$

$$951 = 1667_8$$

$$237_{10} = ? N_2$$

$$237 \xrightarrow[\text{Division sucesiva}]{10 \rightarrow 2} 11101101_2$$



$$11101101_2$$

\* Conclusión: - El método puente es el caso general

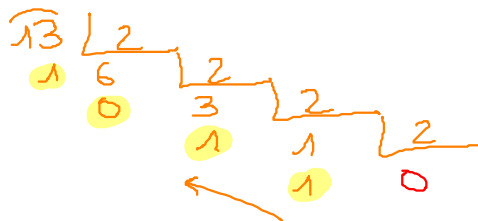
$$15_8 = N_2$$

① Expansión polinómica ( $8 \rightarrow 10$ )

$$\begin{aligned} 15_8 &= 1 \times 8^1 + 5 \times 8^0 \\ &= 1 \times 8 + 5 \times 1 \\ &= 8 + 5 \\ &= 13_{10} \end{aligned}$$

$$15_8 \xrightarrow{8 \rightarrow 10} 13_{10}$$

② Divisiones sucesivas ( $10 \rightarrow 2$ )



$$13_{10} \rightarrow 1101_2$$

$$\text{Rta: } 15_8 \xrightarrow{8 \rightarrow 10} 13_{10} \xrightarrow{10 \rightarrow 2} 1101_2$$

$$15_8 = 1101_2$$

\* Atajo

- Bases 2, 8, 16, ...,  $2^n$  (Atajos)



$8 = 2^3$

| Octal | Binario |
|-------|---------|
| 0     | 000     |
| 1     | 001     |
| 2     | 010     |
| 3     | 011     |
| 4     | 100     |
| 5     | 101     |
| 6     | 110     |
| 7     | 111     |

$2^4$

| Hexadecimal | Binario |
|-------------|---------|
| 0           | 0000    |
| 1           | 0001    |
| 2           | 0010    |
| 3           | 0011    |
| 4           | 0100    |
| 5           | 0101    |
| 6           | 0110    |
| 7           | 0111    |
| 8           | 1000    |
| 9           | 1001    |
| A           | 1010    |
| B           | 1011    |
| C           | 1100    |
| D           | 1101    |
| E           | 1110    |
| F           | 1111    |

$$10 = 00010$$

Ejemplo: Convertir  $11110101101_2$  a octal y hexadecimal

$$\begin{array}{ccccccc} 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline & 3 & 6 & 5 & 5 & & & & & & & \end{array} \quad \begin{array}{l} \leftarrow \\ \text{?} \\ N_8 \end{array}$$

$$= 3655_8$$

$$\begin{array}{ccccccc} 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ \hline & 7 & & A & & & D & & & & & \end{array} \quad \begin{array}{l} \leftarrow \\ \text{?} \\ N_{16} \end{array}$$

$$= 7AD_{16}$$

Convertir  $9F2_{16}$  a binario  $9F2_{16} \stackrel{?}{=} N_2$

$$9F2_{16} = \frac{1001}{9} \frac{1111}{F} \frac{0010}{2} = 100111110010_2$$

Convertir  $170_8$  a binario  $170_8 \stackrel{?}{=} N_2$

$$170_8 = \frac{001}{1} \frac{111}{7} \frac{000}{0} = \cancel{00}1111000_2$$

$$= 1111000_2$$

$2AF_{16} \stackrel{?}{=} N_8$  (Punteo 2)

$$2AF_{16} = \begin{array}{ccccccc} & 1 & & 2 & & 5 & & 7 \\ & \circ & \circ & \circ & \circ & \circ & \circ & \circ \\ \hline & 2 & & A & & & & F \end{array} \quad \begin{array}{l} \leftarrow \\ \text{?} \\ N_8 \end{array}$$

$$= 1257_8$$